

Kaustubh Gupta

AI Engineer | Generative AI • Multi-Agent Systems • RAG • FastAPI

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PROFESSIONAL SUMMARY

Results-driven AI Engineer building scalable Generative AI solutions, multi-agent workflows, and production backend architectures in the BFSI domain. Experienced in enterprise-grade Retrieval-Augmented Generation (RAG) pipelines, event-driven automation, and high-performance microservices. Reduced LLM token consumption by 70% and audit latency by 4x with delta-based document processing, and built a hybrid OFAC/PEP screening engine over 800K+ records. IEEE-published researcher.

TECHNICAL SKILLS

Generative AI & Agents	LangGraph, LangChain, Multi-Agent Systems, Agentic Workflows, LLM Orchestration, RAG, Prompt Engineering, Axolotl (Fine-tuning), QLoRA, Hybrid Search (Dense + BM25), Reciprocal Rank Fusion (RRF), Cross-Encoder Reranking, HyDE, Multi-Query Retrieval, Semantic Chunking, Text-to-SQL, Human-in-the-Loop (HITL), Guardrails, MCP, FastMCP, Pydantic, Playwright, OCR, Structured Extraction
LLMOps & Evaluation	Langfuse (Tracing, Prompt Versioning, Automated Evaluation), RAGAS, MLflow
Vector Search & Retrieval	FAISS, Pgvector, Annoy, BM25, Semantic Search, MinHash De-duplication, Fuzzy Matching (RapidFuzz)
ML & DL	ANN Optimizations, Feature Engineering, CNN, SVM, Decision Trees, Regression, Classification, Clustering, Ensemble Methods, SMOTE, TensorFlow, Hugging Face, Scikit-learn, NumPy, Pandas, Seaborn, SHAP
Backend & Databases	FastAPI, REST APIs, Microservices, Event-Driven Architecture, RabbitMQ, Celery, Ray, Redis, PostgreSQL, DuckDB, Neo4j, Cypher, Knowledge Graphs, Change Data Capture (CDC), SQLAlchemy
Frontend	ReactJS, Zustand, TypeScript, Tailwind CSS, HTML
Deployment & DevOps	Docker, AWS, Infisical, Keycloak, Dokploy, CI/CD, Git
Languages	Python, JavaScript, SQL, TypeScript

PROFESSIONAL EXPERIENCE

Tata Consultancy Services (BFSI R&I Team)

Apr 2025 – Present

AI Engineer

Loan Prebooking System

FastAPI, LangGraph, Langfuse, RabbitMQ, Celery

- Developed an event-driven loan onboarding and underwriting pipeline using LangGraph, automating emailed document parsing, customer record extraction, and regulatory verification to reduce manual processing time by 5x.
- Engineered a multi-agent validation layer with Human-in-the-Loop (HITL) review gates to inspect discrepancy flags, approve edge-case applications, and autonomously compile final compliant loan booking sheets.
- Integrated self-hosted Langfuse for distributed trace logging, latency tracking, prompt versioning, and automated evaluation suites across multi-turn agent runs.

KYC Scrutiny Engine

LangGraph, Celery, DuckDB, Neo4j, RapidFuzz

- Architected a fault-tolerant, stateful KYC scrutiny pipeline using Celery + RabbitMQ and LangGraph persistent workflows to process KYC cases via asynchronous document queues through OCR, structured extraction, and compliance validation rules, reducing processing time from ~15 mins to ~3 mins.
- Implemented multi-hop entity relationship graph traversal in Neo4j for complex ownership and affiliation analysis.
- Crafted a hybrid screening engine combining DuckDB fuzzy pre-filtering with RapidFuzz and semantic LLM matching for high-accuracy OFAC/PEP screening against 800K+ records, with dual-stage Human-in-the-Loop (HITL) checkpoints and full audit trail generation.

Trade Exception Settlement System

LangGraph, Neo4j, CDC, DuckDB, Guardrails, Text-to-SQL

- Built an autonomous exception settlement workflow using LangGraph-based orchestration, running Cypher filtering over historical playbooks to retrieve past resolutions and verifying counterparty & SSI data to resolve trade discrepancies and operational breakages for trade ops teams, reducing exception resolution time by ~60%.
- Used a Neo4j Knowledge Graph to visualize end-to-end trade lifecycles and downstream cascade dependencies.

- Implemented Change Data Capture (CDC) pipelines to stream live updates from PostgreSQL into DuckDB in real time, cutting analytics data latency by ~30% versus batch refresh and powering a secured Text-to-SQL chatbot with robust Guardrails for post-trade analytics dashboards.
- Configured agentic email dispatch to autonomously draft and route context-aware discrepancy resolution notices to counterparties and internal operations desks.

Regulatory Compliance Check

LangChain, RAGAS, FAISS, FastAPI, Ray

- Developed a compliance audit framework for 150+ page policy documents with a Git-style document versioning algorithm that processes only modified clauses (deltas), reducing LLM token consumption by 70% and audit latency by 4x.
- Optimized large-scale retrieval with MinHash-based chunk de-duplication and semantic chunking through a local embedding model, improving chunking quality and speed.
- Engineered an advanced hybrid RAG architecture combining dense vector embeddings with sparse BM25 retrieval, optimized via Reciprocal Rank Fusion (RRF) and cross-encoder reranking for precise clause extraction.
- Integrated automated RAG evaluation pipelines using RAGAS (faithfulness, answer relevancy, context precision) and applied HyDE & multi-query retrieval for global policy search based on user queries.
- Crafted parallel clause-level PDF highlighting using Ray to distribute CPU-bound tasks across available cores (3x faster processing), and rendered visual document topic flows using Mermaid.js.

KEY PROJECTS

Transcribe – Lecture Transcription Workspace

[Live Demo](#)

React.js, FastAPI, RabbitMQ, Celery, PostgreSQL, FFmpeg, Gemini

- Developed an AI-powered lecture transcription platform that converts long-form spiritual discourses into structured, searchable, and downloadable PDF transcripts.
- Built a responsive system for authentication, file uploads, transcription history, live processing status through SSE, and document downloads.
- Implemented an asynchronous processing pipeline using Celery workers and RabbitMQ queues, FFmpeg-based media conversion, silence detection, and chunked processing for scalable long-form transcription.
- Optimized transcription quality for Sanskrit shlokas (verses) and multilingual pravachans (discourses) using Gemini, followed by AI-powered transcript aggregation into structured HTML and automated PDF generation for 50+ active users.

Two-Tower Recommendation System

March 2024

Python, Kaggle, Hugging Face, TensorFlow, Annoy, Pandas

- Architected a deep Two-Tower retrieval network mapping queries and products into a shared high-dimensional vector space to deliver personalized recommendations.
- Engineered a Product Tower that concatenates diverse feature embeddings (title, description, bullet-points, brand, and color) before passing them through dedicated dense processing layers.
- Implemented L2-Normalization on output layers to stabilize dot-product similarity scoring, optimizing the network with Cross-Entropy Loss on an Amazon dataset of 150K+ records.
- Accelerated nearest-neighbor retrieval by indexing the fine-tuned representations with Annoy, enabling scalable, low-latency item matching.

KEY ACHIEVEMENTS

LLM Cost & Latency	70% fewer LLM tokens and 4x lower audit latency via Git-style delta processing
Processing Speed	3x faster clause-level PDF highlighting by parallelizing CPU-bound tasks with Ray
Screening at Scale	Hybrid OFAC/PEP screening (DuckDB + RapidFuzz + LLM matching) across 800K+ records
Resolution & Data Speed	~60% faster trade exception resolution; ~30% lower analytics data latency via CDC
Production Adoption	Lecture transcription platform serving 50+ active users
Predictive Accuracy	90% churn-prediction accuracy with ANN, Random Forest, AdaBoost soft-voting ensemble (BCG, Forage)
Publication	IEEE INDIACom, Mar 2024: Binary Classification on Students' Dropout; DOI: 10.23919/INDIACom61295.2024.10498546

EDUCATION

KIET Group of Institutions, Delhi-NCR

2020 – 2024

B.Tech in CS & IT | CGPA: 8.04 / 10.0

Certifications: AWS Academy Cloud Foundations (AWS Academy, Mar 2024); BCG Data Science Job Simulation (Forage, Mar 2024)